

KOUSHIK GUDIPELLY

Backend / Python Software Engineer

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SUMMARY

Backend-focused Software Engineer with **4+ years** of experience designing and shipping high-throughput microservices, event-driven data pipelines, and distributed systems at Bank of America and Deloitte. Deep expertise in **Python, FastAPI, Apache Kafka, and Spark**, building APIs and services that process **millions of transactions daily** with sub-second latency. Proven track record cutting query times, shrinking deployment cycles, and shipping ML-backed backend services. Comfortable operating end-to-end across backend, full-stack, and ML/AI engineering roles.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL

Backend & APIs: FastAPI, Flask, Spring Boot, REST APIs, GraphQL, WebSockets, Microservices, Async Programming

Data & Streaming: Apache Kafka, Spark Streaming, PySpark, ETL Pipelines, Event-Driven Architecture

Cloud & DevOps: Azure (AKS, App Services), AWS (EC2, S3, RDS, Glue, Cognito), Docker, Kubernetes, Jenkins, GitLab CI/CD

Databases: PostgreSQL, MySQL, Redis, Snowflake

ML & Analytics: XGBoost, Scikit-learn, TensorFlow, Pandas, NumPy, Feature Engineering, Anomaly Detection, CNNs

Monitoring & Security: Prometheus, Grafana, JWT, OAuth2, RBAC, SonarQube

Testing: pytest, Mockito, Unit & Integration Testing

EXPERIENCE

Software Engineer — Backend / Python | Bank of America — NC, USA

Aug 2024 – Present

- Architected a real-time fraud detection backend processing **1–2M daily transactions** using FastAPI, Python, Apache Kafka, and Spark Streaming, achieving sub-second fraud scoring latency.
- Built and deployed ML-backed fraud scoring services (XGBoost, Scikit-learn) as FastAPI microservices, improving fraud detection accuracy by **12–15%** and reducing false positives.
- Designed scalable Kafka and Spark data pipelines for real-time ingestion, transformation, and validation of high-volume financial transaction streams.
- Developed rule-based fraud detection logic (velocity checks, geo-location anomalies, device fingerprinting, behavioral profiling), increasing early detection effectiveness by **15%**.
- Optimized MySQL schema design, indexing strategy, and query execution, reducing query times by **20%** across core transaction tables.
- Automated CI/CD pipelines with Jenkins, Docker, and Kubernetes (SonarQube + Nexus), cutting deployment cycle time by **30%** and improving release consistency.
- Implemented JWT-based authentication and RBAC across microservices, ensuring secure access control and financial compliance standards.

Full Stack Software Developer | Deloitte — Hyderabad, India

Jan 2021 – Jul 2023

- Designed and delivered a Python/FastAPI microservices backend for an e-commerce order management platform handling **40K–50K daily transactions**, improving system stability under peak load.
- Led migration of a legacy monolithic system to domain-driven microservices (FastAPI async + Spring Boot), reducing release cycles by **30%** and enabling independent service deployments.
- Built RESTful and GraphQL APIs with FastAPI, Pydantic, and OpenAPI, improving response latency from ~300ms to **~200ms** with strict schema validation.
- Implemented event-driven order tracking using Apache Kafka and WebSockets, reducing processing lag during peak traffic by **25%**.
- Optimized backend performance via Redis caching and async processing, cutting API response latency from ~500ms to **~300ms**.
- Established CI/CD pipelines with GitLab CI, Docker, and pytest, improving build reliability, code quality, and deployment consistency.
- Deployed and monitored cloud-native backend services on Azure AKS with Prometheus and Grafana observability, maintaining high availability in production.

PROJECTS

AI Resume Analyzer | *Generative AI, FastAPI, React, OpenAI APIs*

- Built a workflow automation tool using FastAPI, React, OpenAI APIs, and LLM workflows to analyze resumes and job descriptions.
- Designed prompt and retrieval workflows for scoring, keyword matching, and recommendation generation.
- Implemented secure REST APIs, validation, logging, and reusable backend components for human-facing AI feedback.

Project Management App | *Next.js, React, TypeScript, Node.js, AWS*

- Built a full-stack project management platform using Next.js, React, TypeScript, Node.js/Express, Prisma, and typed REST APIs.
- Designed AWS deployment using EC2, RDS, Amplify, and S3 with Cognito-based authentication.
- Implemented dashboards, timelines, Gantt views, and analytics workflows for product tracking.

EDUCATION

Master of Science in Computer Science
NC State University, Raleigh, NC, USA

Aug 2023 – May 2025